

Introducing Flutter

Google's new mobile application development SDK

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Have you ever wondered how easy things might have been if mobile developers could build applications for several platforms from just one single code base? Fascinating isn't it? Developers could just dedicate their time to focus on the code they want to create or the logic they intend to build without letting their mind be clouded by the impositions of environment/operating system on which the app shall run. Well, worry no more for Google has yet again come forward with a boon for the developers in the form of Flutter.

Flutter allows you to build beautiful, native apps on iOS and Android from a singular codebase. In other words, the code that you write once shall be taken care of by Flutter to suit both Android and iOS, two of the most widely used mobile operating systems in the current times.

Google claims the four tenets of Flutter to be beautiful, fast, productive and open. To have an understanding of how fast Flutter can be, it's good to know that Flutter compiles directly to ARM code without any scripts to parse. To justify its openness, Flutter is not just open source but it allows us to build on top of applications and frameworks on it.

One of the most amazing aspects of Flutter is the functionality of 'Hot Reload' in which all you need to do is to save the app and hit refresh to let the modified code work its way and dis-

play the changes made to the app itself in a matter of few milliseconds which can almost be thought of as refreshing a webpage. Imagine the amount of time that can be saved and also the difference this brings to the usual way of modifying code to see the changes implemented where several files had to be recompiled.

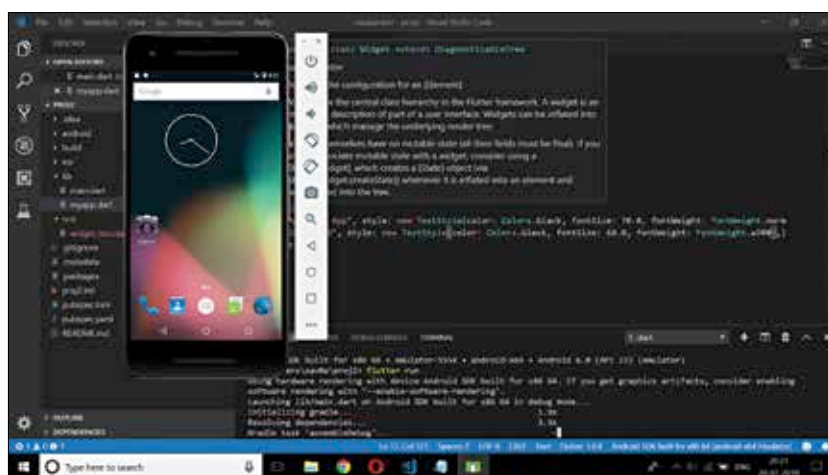
Flutter allows building vector animations at every frame which aren't essentially mistaken to be pre-computed gifs and they run at 60 frames per second. At its core, Flutter uses DART as its programming language,

stateful widget retains state information and a stateless widget does not.

With this insight on Flutter, let's have it installed on our computers. It is a fairly easy process for both macOS and Windows users with a slight snag for the latter to which we shall get into steadily. We'll discuss the process for Windows here. For macOS, it's even easier than the Windows one..

Things you'll need

The first thing that we shall need is our very own command prompt terminal or a suitable alternative to that,



The emulator running on Google Pixel 2 running Android 7.1

which was also developed by Google a long time back. It didn't receive much popularity back then but it's back into relevance with the advent of Flutter.

Flutter uses widgets, which are the basic building blocks of most web or mobile app these days. They comprise the UI elements that we see on screen and offer minute or specific controls inside the app. Flutter provides for stateful and stateless widgets. According to Flutter, a stateless widget is one that does not involve a change in its state whereas a stateful widget maintains information of the state which are subject to change. Essentially, a

preferably the git bash for executing certain flutter commands which aid this process. We also need to get the Java SDK installed and set the path to the bin folder of the directory in which Java was installed. We then need to consider whether we would be testing out our Flutter code in a real Android device (a smartphone) or a virtual emulator such as the one that the Android Studio provides. In case of the latter, we need to install Android Studio which sets up an emulator for Android. The emulator used for building a small application has been run on a Google Pixel 2 running on